

Laboratory Report

EIT & MEC – Digi Lab 1: Switching Circuits

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Abstract

This report documents the experimental investigation of monostable and astable multivibrators, conducted in Digi Lab 1 at Fachhochschule Vorarlberg on 21/22 April 2026. Three circuits were assembled and analysed: a monostable multivibrator based on the NE555 timer IC, an astable multivibrator also using the NE555, and an astable oscillator built from two CMOS inverters (74C00). Calculated values for pulse duration and oscillation period were compared against oscilloscope measurements. The monostable circuit showed a deviation of 7.3%, increasing to 8.2% with a second IC. Direct BNC measurements proved that low-impedance connections disrupt the circuit, preventing the output from returning to low. The astable NE555 circuit oscillated at 495.8 Hz, deviating 3.1% from theory. Results confirmed that component tolerances and probe loading are the primary causes of error. Observations also showed that bypass capacitors are essential for stable switching edges. All circuits performed according to theoretical models within acceptable margins.

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1 Introduction

Switching circuits are fundamental building blocks in digital and mixed-signal electronics. This laboratory exercise focuses on two important classes of such circuits: multivibrators, which generate or shape rectangular pulse signals, and oscillators built from logic gates.

A **monostable multivibrator** (one-shot) produces a single output pulse of defined duration in response to a trigger event. An **astable multivibrator** oscillates continuously between two states without any external trigger, generating a periodic square wave. Both types can be realised with the NE555 timer IC, whose internal structure consists of two comparators, a flip-flop, a discharge transistor, and an output stage.

The **NE555** is a versatile and widely used timer IC. In monostable mode, the output pulse width is determined solely by an external RC network according to $t_H = 1.1 RC$. In astable mode, the frequency and duty cycle depend on two resistors and one capacitor.

As a complementary approach, an astable oscillator can also be constructed from two cascaded CMOS inverter gates with a feedback RC network.

The objectives of this laboratory are:

- To understand and verify the operating principle of monostable and astable multivibrators using the NE555 IC.
- To compare calculated and measured values for pulse duration and oscillation period, and to quantify and explain deviations.
- To investigate the influence of the measurement setup (oscilloscope probes vs. BNC cables) on the circuit behaviour.
- To analyse the astable multivibrator realised with two CMOS inverters.

The theoretical background was reviewed according to [1] and [2].

2 Equipment and Materials

All circuits were assembled on a standard experimentation (breadboard) system. Signals were observed and recorded with a digital storage oscilloscope. Table 1 lists the equipment used.

Table 1: Equipment and components used in this experiment.

Item	Specification / Part No.	Asset / S/N
Digital oscilloscope	Teledyne LeCroy HD 4096	—
Oscilloscope probe (10:1)	—	—
BNC cable	—	—
DC power supply	10 V	—
Experimentation board	—	—
NE555 timer IC	NE555	—
CMOS NAND IC	74C00	—
Resistors	see individual tasks	—
Capacitors	see individual tasks	—

3 Theoretical Background

3.1 NE555 Internal Structure

The NE555 contains three equal resistors forming a voltage divider that sets the reference levels at $\frac{1}{3}V_{CC}$ and $\frac{2}{3}V_{CC}$, two comparators, an SR flip-flop, an npn discharge transistor (pin 7), and a push-pull output stage (pin 3).

3.2 Monostable Mode

In monostable mode, the capacitor C charges through R_1 after a falling trigger edge at pin 2 pulls the output high. The output returns low once V_C reaches $\frac{2}{3}V_{CC}$, which gives the pulse duration:

$$t_H = 1.1 \cdot R_1 \cdot C \quad (1)$$

3.3 Astable Mode

In astable mode, C charges through $R_1 + R_2$ and discharges through R_2 alone. The resulting high and low times are:

$$t_H = 0.693 (R_1 + R_2) C \quad (2)$$

$$t_L = 0.693 R_2 C \quad (3)$$

$$T = 0.693 (R_1 + 2R_2) C \quad (4)$$

3.4 RC Oscillator with CMOS Inverters

Two cascaded inverters with a series resistor R and capacitor C in the feedback path form a relaxation oscillator. The threshold voltages of the CMOS gates determine the switching points, giving an oscillation frequency of approximately:

$$f \approx \frac{1}{2.2 RC} \quad (5)$$

4 Procedures

All experiments were carried out according to the laboratory instruction sheet *Lab_01_SwitchingCircuits*, Rev. 1.1, dated 21/22.04.2026. Circuits were assembled on the experimentation board. A supply voltage of 10 V was used for all tasks.

Oscilloscope measurements were performed in two configurations where applicable:

1. Using 10:1 oscilloscope probes (input impedance 10 M Ω).
2. Using direct BNC cables (1 M Ω input impedance, task 3.1.4 only).

Screenshots were saved for each relevant waveform. Component values were verified with a multimeter before assembly where possible.

5 Results and Discussion

5.1 Task 3.1 – Monostable Multivibrator (NE555)

Component values used: $C = 1 \mu\text{F}$, $R_1 = 1 \text{ M}\Omega$, $V_{CC} = 10 \text{ V}$.

5.1.1 Task 3.1.1 – Calculation of Pulse Duration

Applying equation (1):

$$\begin{aligned} t_{H,\text{calc}} &= 1.1 \times 1 \text{ M}\Omega \times 1 \mu\text{F} \\ &= 1.1 \text{ s} \end{aligned}$$

$$t_{H,\text{calc}} = 1.1 \text{ s}$$

5.1.2 Task 3.1.2 – Measurement with Oscilloscope Probes (10:1)

Both the output signal y (pin 3) and the capacitor voltage V_C were measured simultaneously using 10:1 oscilloscope probes. The measurement was carried out on two different NE555 ICs under identical conditions.

Table 2: Measured values – monostable multivibrator, 10:1 oscilloscope probes.

Quantity	Symbol	Calculated	Measured
Pulse duration (NE555 #1)	t_H	1.1 s	1.18 s
Pulse duration (NE555 #2)	t_H	1.1 s	1.19 s
Peak capacitor voltage	$V_{C,\text{max}}$	6.67 V	6.8 V

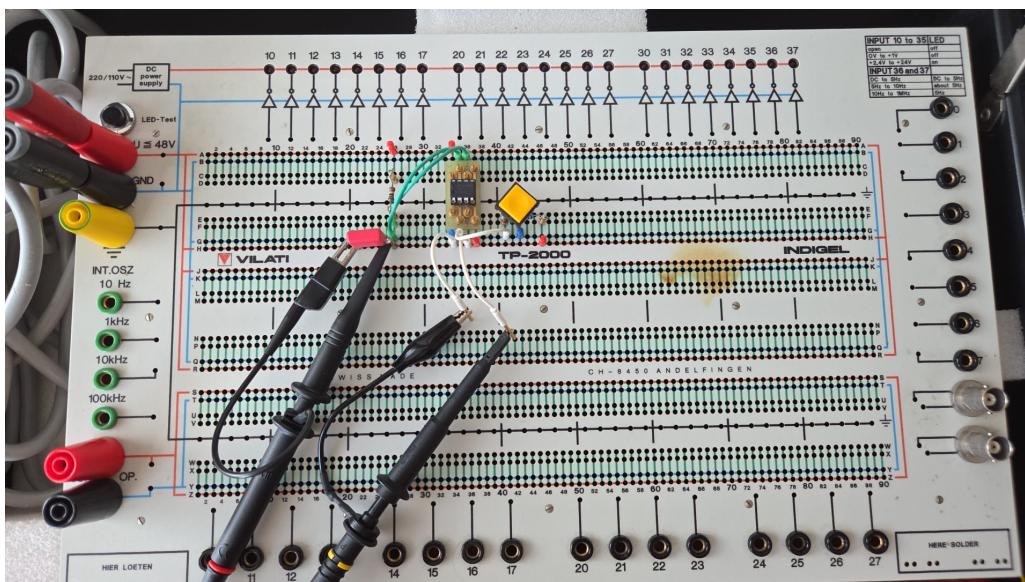


Figure 1: Measuring setup

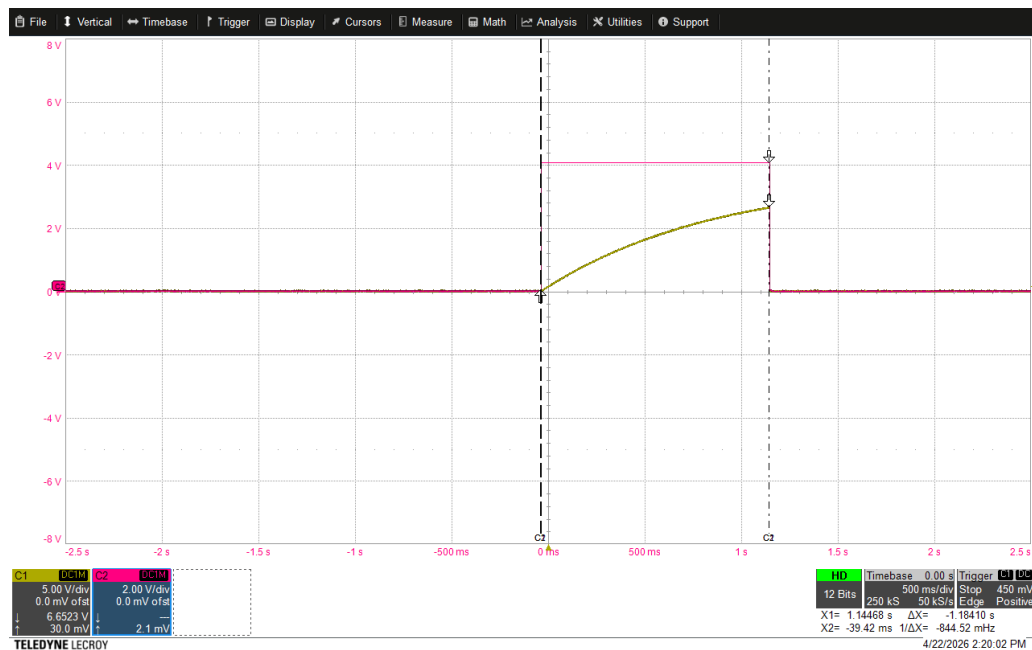


Figure 2: Output y (C2 - red trace) and capacitor voltage V_C (C1 - yellow trace) measured with 10:1 probes – monostable multivibrator - Erster NE555

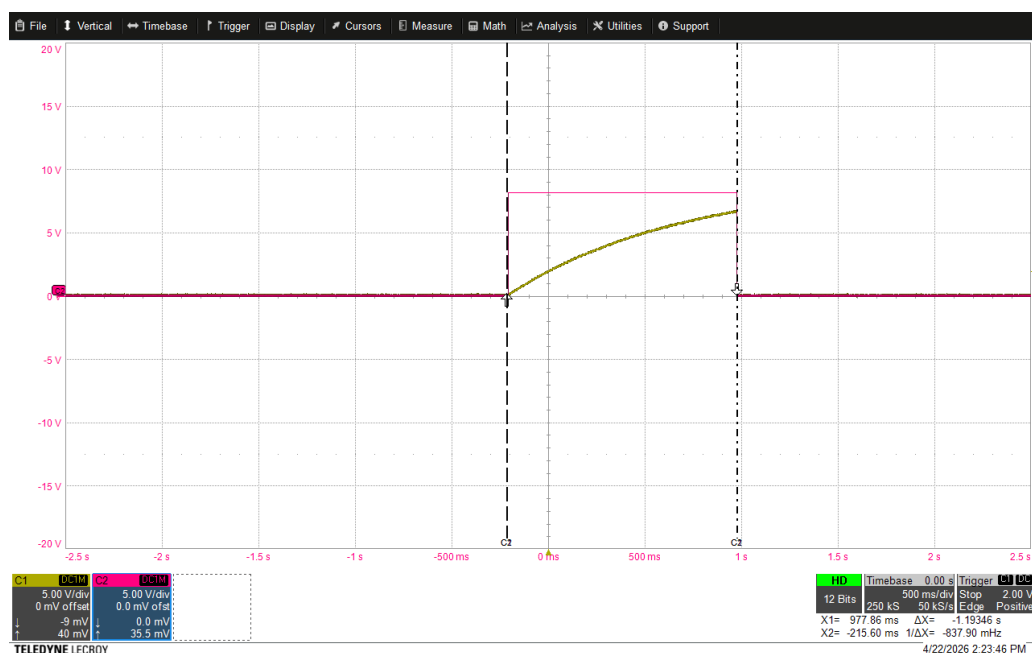


Figure 3: Output y (C2 - red trace) and capacitor voltage V_C (C1 - yellow trace) measured with 10:1 probes – monostable multivibrator - Zweiter NE555

5.1.3 Task 3.1.3 – Explanation of Differences

The measured pulse durations of 1.18 s and 1.19 s deviate from the calculated value of 1.1 s by approximately 7.3 % and 8.2 % respectively. The measured capacitor peak voltage of 6.8 V is slightly above the theoretical threshold of $\frac{2}{3} \cdot 10 \text{ V} = 6.67 \text{ V}$.

The primary cause of the deviation is the **capacitor tolerance** of $\pm 20 \%$, which alone is sufficient to explain the observed discrepancy. Additionally, the 10:1 probe introduces a

parasitic capacitance of approximately 10 pF in parallel with the timing capacitor, which slightly increases the effective capacitance and therefore the pulse duration. The NE555 internal comparator threshold also deviates slightly from the ideal $\frac{2}{3}V_{CC}$, contributing a small systematic offset.

5.1.4 Task 3.1.4 – Measurement with BNC Cables

The measurement from task 3.1.2 was repeated, replacing the 10:1 oscilloscope probes with direct BNC cables.

A BNC cable connected to the oscilloscope presents an input impedance of 1 M Ω , which is placed **in parallel with the timing capacitor**. Together with the series resistance $R_1 = 1\text{ M}\Omega$ of the circuit, this forms a voltage divider with a ratio of:

$$\frac{R_{\text{BNC}}}{R_1 + R_{\text{BNC}}} = \frac{1\text{ M}\Omega}{1\text{ M}\Omega + 1\text{ M}\Omega} = 0.5$$

As a result, the maximum voltage across the capacitor is limited to approximately $0.5 \times 10\text{ V} = 5\text{ V}$, which never reaches the NE555 threshold of $\frac{2}{3}V_{CC} = 6.67\text{ V}$. Consequently, the comparator is never triggered and the output **never returns to low** – the monostable behaviour is completely disrupted.

In contrast, the 10:1 probe provides 10 M Ω input impedance, which is negligible compared to $R_1 = 1\text{ M}\Omega$ and does not significantly alter the circuit behaviour.

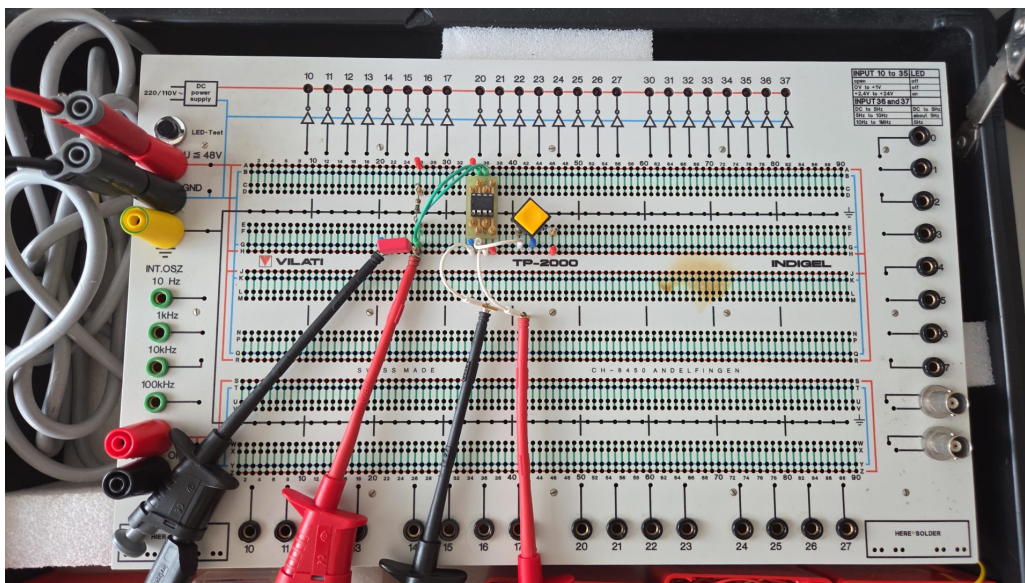


Figure 4: Measuring setup

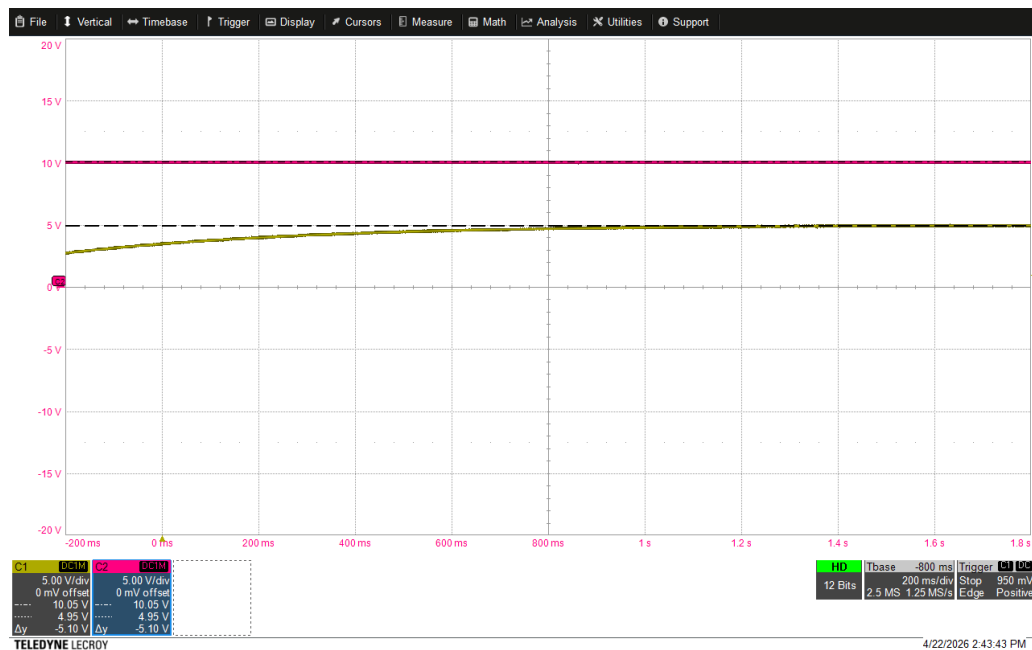


Figure 5: Output y (C2 - red trace) and V_C (C1 - yellow trace) measured via BNC cable – monostable multivibrator. The output does not return to low due to voltage divider loading.

5.1.5 Task 3.1.5 – Output-Only Measurement (No Capacitor Probing)

The output signal y at pin 3 was measured alone, without a probe connected to the capacitor node.

Table 3: Output pulse duration with and without simultaneous capacitor probing.

Quantity	With V_C probed	Without V_C probed
Pulse duration t_H	1.18 s	1.08 s

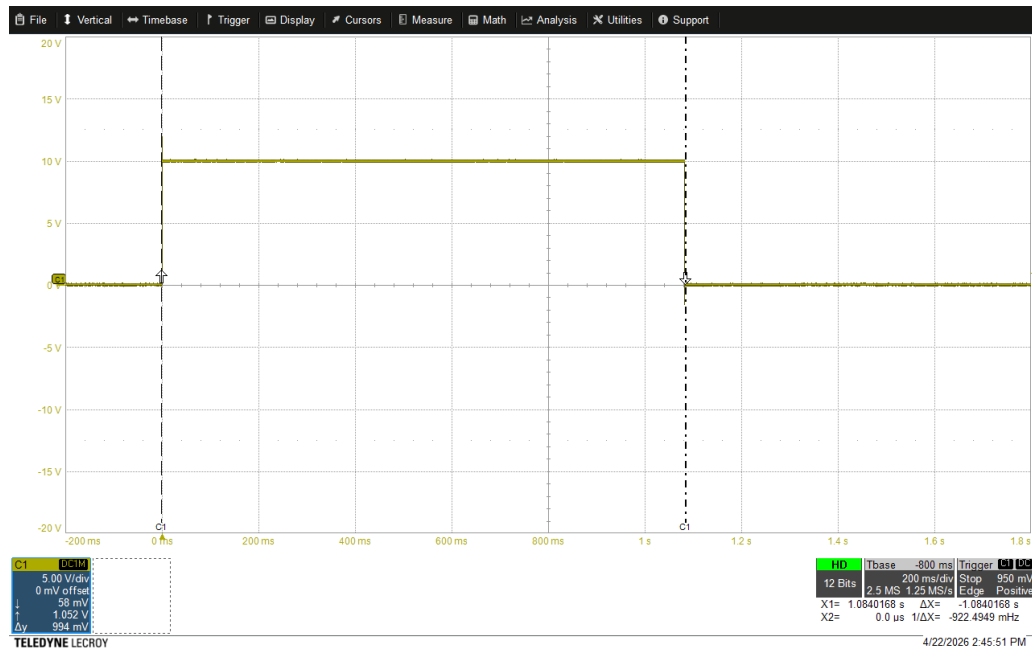


Figure 6: Output y (C1 - yellow trace) only – monostable multivibrator, no capacitor probing.

When the probe is connected to the capacitor node, a small current flows through the probe's input resistance ($10\text{ M}\Omega$) in parallel with the capacitor. This additional current path **partially discharges the capacitor**, meaning the capacitor must charge for a longer time to reach the NE555 threshold voltage. Therefore, the pulse duration is longer when the capacitor is probed (1.18 s) compared to unprobed (1.08 s). The difference of 0.1 s ($\approx 8.5\%$) confirms the measurable but not destructive loading effect of the 10:1 probe on a high-impedance node.

5.2 Task 3.2 – Astable Multivibrator (NE555)

Component values used: $R_1 = 10\text{ k}\Omega$, $R_2 = 10\text{ k}\Omega$, $C = 100\text{ nF}$, $V_{CC} = 10\text{ V}$.

5.2.1 Task 3.2.1 – Calculation of Oscillation Period

Applying equations (2)–(4):

$$t_H = 0.693 \times (10\text{ k}\Omega + 10\text{ k}\Omega) \times 100\text{ nF} = 1.386\text{ ms}$$

$$t_L = 0.693 \times 10\text{ k}\Omega \times 100\text{ nF} = 0.693\text{ ms}$$

$$T = t_H + t_L = 2.079\text{ ms}$$

$$f = \frac{1}{T} = 481.0\text{ Hz}$$

$$D = \frac{t_H}{T} = \frac{R_1 + R_2}{R_1 + 2R_2} = 66.7\%$$

$$T_{\text{calc}} = 2.079\text{ ms}, \quad f_{\text{calc}} = 481\text{ Hz}, \quad D_{\text{calc}} = 66.7\%$$

5.2.2 Task 3.2.2 – Measurement

Table 4: Measured values – astable multivibrator (NE555).

Quantity	Symbol	Calculated	Measured
High time	t_H	1.386 ms	1.340 ms
Low time	t_L	0.693 ms	0.663 ms
Period	T	2.079 ms	2.017 ms
Frequency	f	481.0 Hz	495.8 Hz
Duty cycle	D	66.7 %	66.4 %

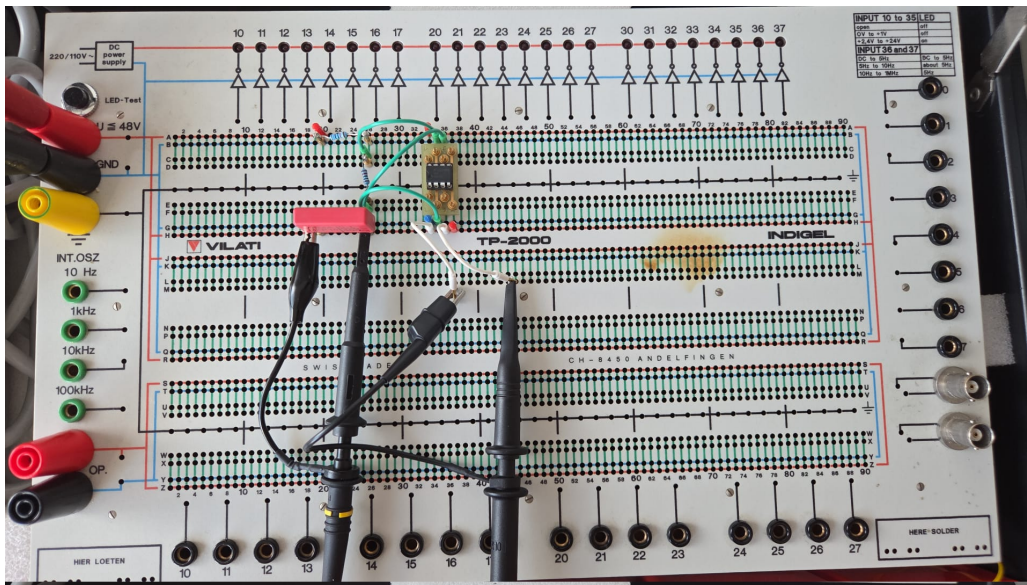


Figure 7: Measuring setup

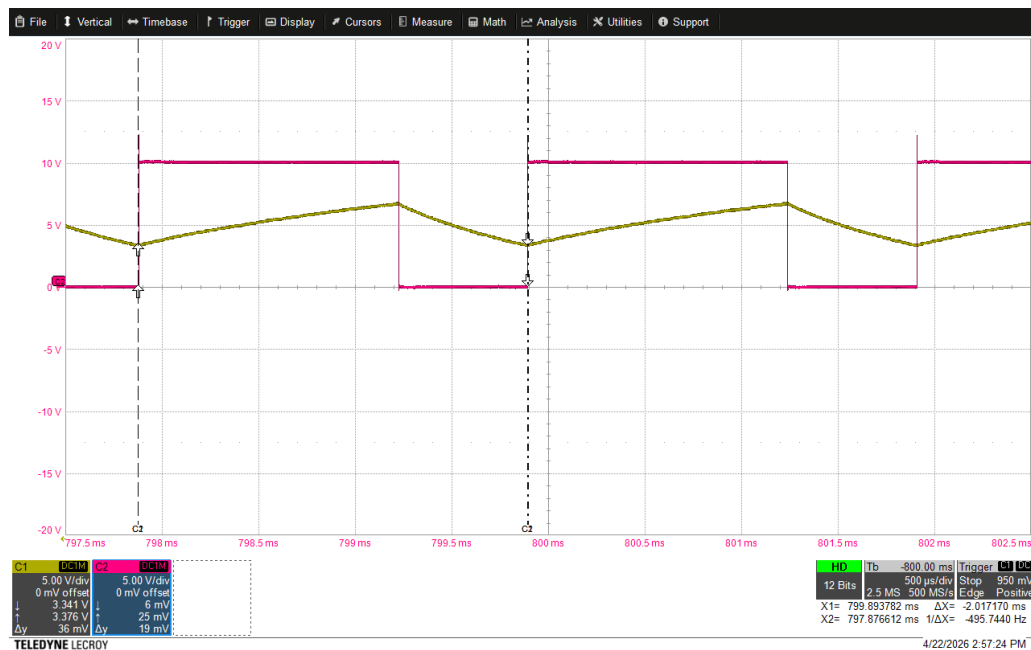


Figure 8: Output y (C2 - red trace) and V_C (C1 - yellow trace) – astable multivibrator, NE555.

5.2.3 Task 3.2.3 – Explanation of Differences

The measured period of 2.017 ms deviates from the calculated 2.079 ms by approximately 3.0%. This deviation is well within the expected component tolerances of $\pm 5\%$ for resistors and $\pm 20\%$ for capacitors and can therefore be attributed primarily to **component tolerances**.

Compared to task 3.1, the influence of the oscilloscope probe on the measurement is significantly smaller. The probe's $10\text{ M}\Omega$ input impedance is negligible relative to the circuit resistances of only $10\text{ k}\Omega$, whereas in task 3.1 the probe impedance was of the same order of magnitude as $R_1 = 1\text{ M}\Omega$. Consequently, probe loading has a much smaller effect here.

5.2.4 Task 3.2.4 – Rising Edge Zoom

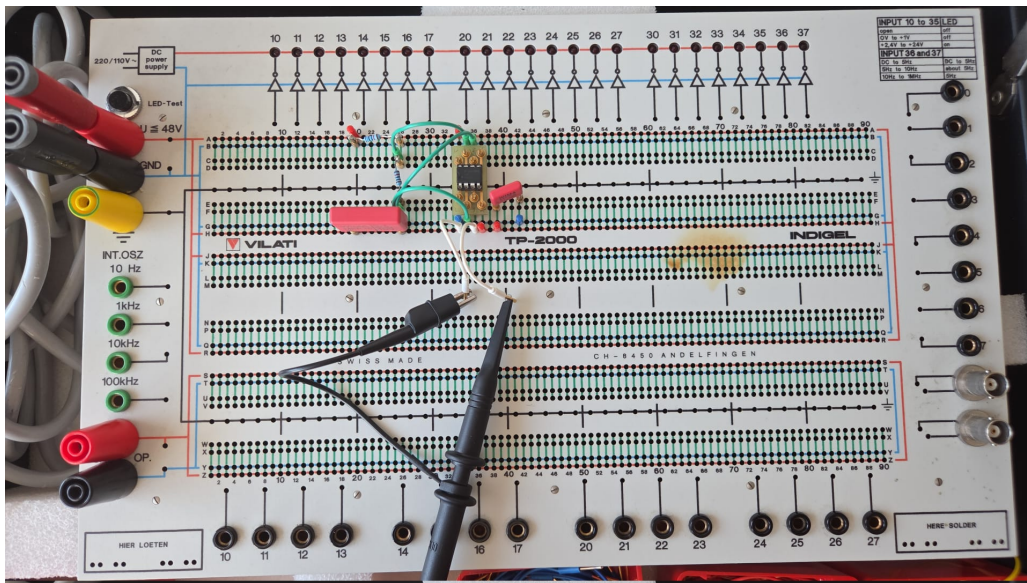


Figure 9: Measuring setup



Figure 10: Zoomed rising edge of the NE555 astable output signal without a bypass (decoupling) capacitor.



Figure 11: Zoomed rising edge of the NE555 astable output signal with a bypass (decoupling) capacitor.

The rising edge of the NE555 output does not exhibit a perfectly steep transition. This is caused by the output stage of the NE555, which consists of a **BJT totem-pole structure**. When the output transistor switches from saturation to cut-off, carrier storage time and the capacitive load at pin 3 (wiring, probe) slow down the transition.

Furthermore, a **bypass (decoupling) capacitor** must be placed between V_{CC} and GND as close as possible to the IC supply pins. When the NE555 output switches, the IC momentarily draws a large transient current. Without a bypass capacitor, this current must be supplied from the power supply through the full inductance of the supply wires, causing a voltage dip at V_{CC} and potential instability of the switching threshold. The bypass capacitor acts as a local energy reservoir that immediately provides the required transient current, ensuring stable and reliable switching behaviour.

5.3 Task 3.3 – Astable Multivibrator Using Two Inverters (74C00)

Component values used: $R_S = 1\text{ k}\Omega$, $R = 10\text{ k}\Omega$, $C = 1\text{ nF}$, $V_{CC} = 10\text{ V}$.

Expected frequency from equation (5):

$$f_{\text{calc}} \approx \frac{1}{2.2 \times 10\text{ k}\Omega \times 1\text{ nF}} = 45.5\text{ kHz}$$

5.3.1 Task 1.1 – Circuit Function

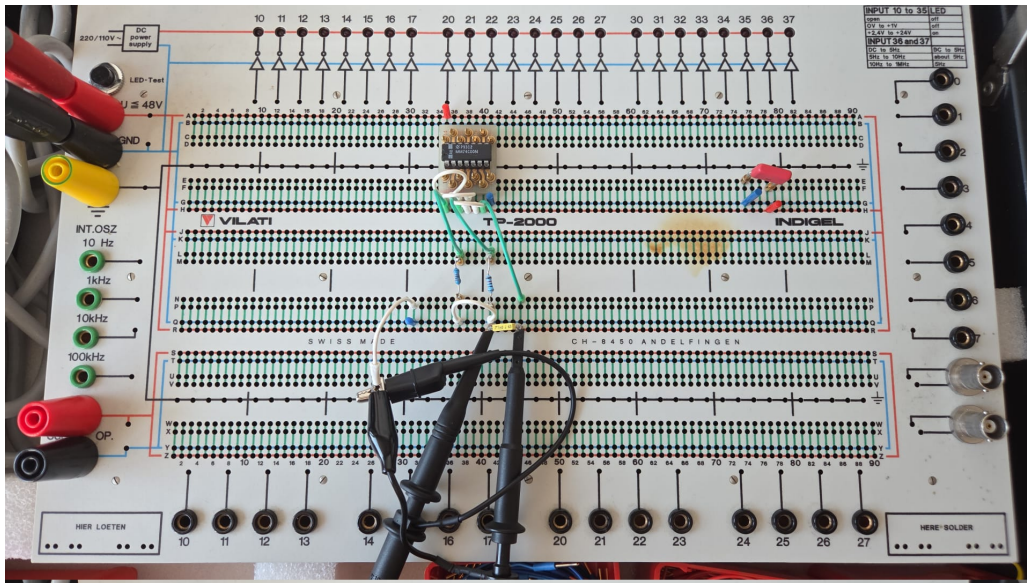


Figure 12: Measuring setup

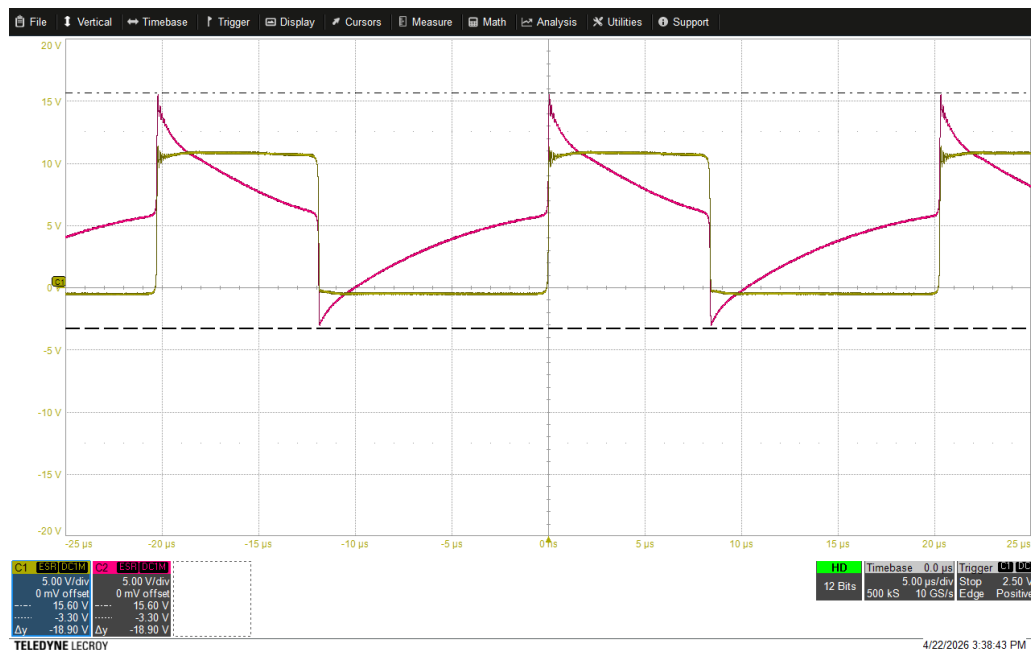


Figure 13: Voltage at node P (C2 - red trace) and output U_{OUT} (C1 - yellow trace) – two-inverter oscillator.

The circuit operates as a relaxation oscillator. The first inverter output feeds back through the RC network (R and C) to its own input at node S. When U_{OUT} switches to a high level, the capacitor C begins charging through R , causing the voltage at node S to rise exponentially. Once V_S crosses the CMOS switching threshold (approximately $\frac{1}{2}V_{CC}$), the first inverter switches its output low, which in turn causes U_{OUT} to go high via the second inverter. The capacitor then discharges in the opposite direction, and the cycle repeats, producing a continuous square wave at the output.

[Describe what is observed on the oscilloscope at U_{OUT} and P. State the observed frequency.]

5.3.2 Task 1.2 – Measurement at S and P

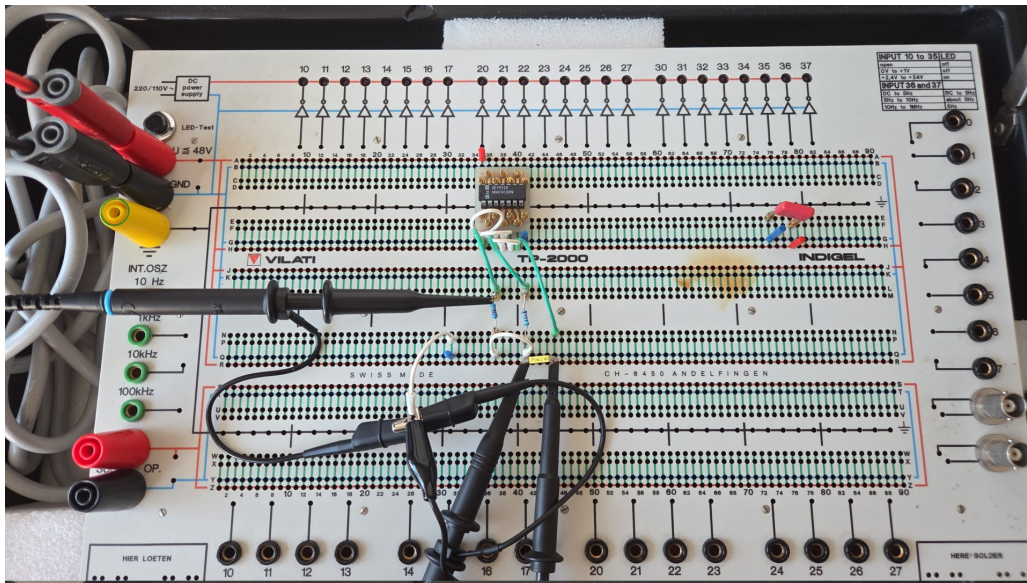


Figure 14: Measuring setup

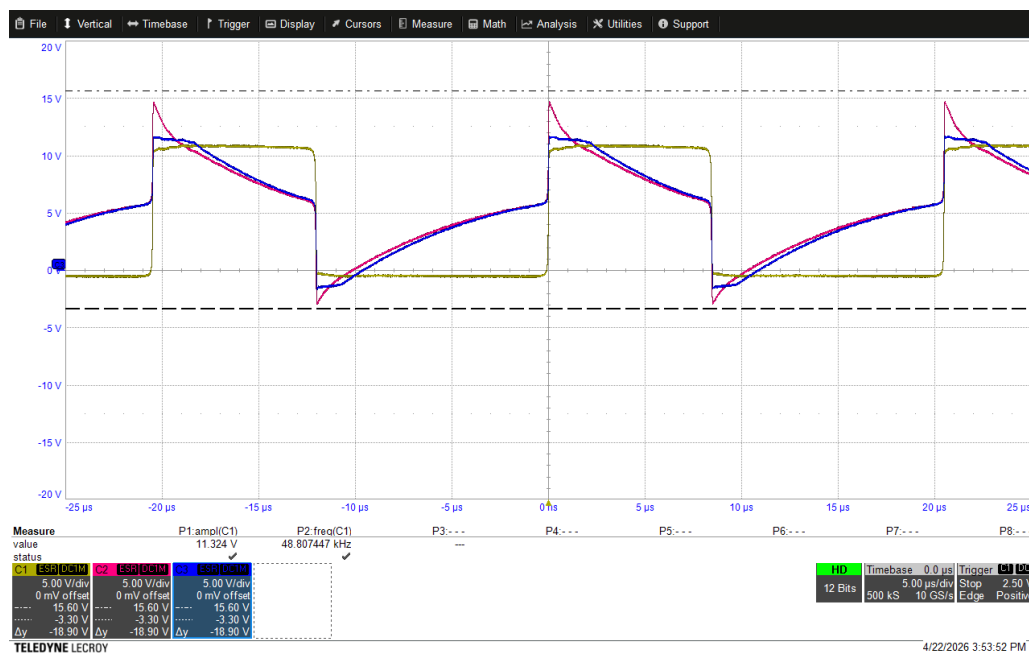


Figure 15: Voltage at node P (C2 - red trace) and output U_{OUT} (C1 - yellow trace) and at node S (C3 - blue trace) – two-inverter oscillator.

At node S, a square wave switching between 0 V and V_{CC} is observed. At node P, an exponential charging and discharging waveform is visible. Due to the RC feedback, the voltage at P swings beyond the supply rails during each transition; it is clamped to approximately -0.7 V and $V_{CC} + 0.7$ V by the internal ESD protection diodes of the CMOS input. This clipping is clearly visible in the oscilloscope trace.

[Fill in the measured values in the table above and describe the waveforms in further detail.]

5.3.3 Task 1.3 – Purpose of R_S

Without R_S , the voltage at node P would exceed the supply rails at every switching event. The resulting current would flow directly through the CMOS input protection diodes without any current limitation, potentially **destroying the IC**.

Additionally, R_S forms part of the total series resistance in the RC network and therefore has a small influence on the oscillation frequency.

6 Conclusions

The following table provides a concise overview of calculated vs. measured values for all tasks:

Table 5: Summary: calculated vs. measured values for all tasks.

Task	Quantity	Calculated	Measured	Rel. error
3.1	t_H (monostable, #1)	1.100 s	1.18 s	7.3 %
3.1	t_H (monostable, #2)	1.100 s	1.19 s	8.2 %
3.2	T (astable NE555)	2.079 ms	2.017 ms	3.0 %
3.2	f (astable NE555)	481 Hz	495.8 Hz	3.1 %
3.3	f (inverter osc.)	45.5 kHz	?? kHz	?? %

All measurements are in good agreement with the theoretical calculations, with deviations well within the expected component tolerances. The largest relative error occurred in task 3.1, where the high impedance of the circuit made it particularly susceptible to probe loading effects. In task 3.2, the probe loading was negligible due to the much lower circuit impedance. The BNC cable measurement in task 3.1.4 demonstrated clearly that a low-impedance measurement connection can fundamentally alter circuit behaviour by acting as a load. The necessity of the bypass capacitor in task 3.2.4 was confirmed by the observed instability in the rising edge without it.

7 Summary

This laboratory exercise verified the timing characteristics of NE555 and CMOS-based switching circuits. Measurements of the monostable multivibrator yielded a pulse duration of 1.18 s, while the astable configuration produced a frequency of 495.8 Hz. These results deviate from theoretical predictions by 7.3% and 3.1%, respectively, primarily due to capacitor tolerances and probe loading. The experiments demonstrated that high-impedance nodes require 10:1 probes to maintain circuit functionality. Ultimately, the data confirms that RC timing models accurately predict behavior when considering practical measurement effects and proper decoupling.

8 Outlook

Future experiments could investigate methods to achieve a 50% duty cycle in the NE555 astable mode by placing a signal diode in parallel with R_2 . This modification allows the capacitor to charge only through R_1 , enabling independent control of high and low times. Another relevant study involves testing the temperature dependence of the RC-based oscillators to quantify frequency drift under thermal stress. Additionally, varying the bypass capacitor values would help determine the minimum capacitance required to maintain switching stability across different frequency ranges.

References

- [1] T. Floyd, *Digital Fundamentals*, 9th ed. Pearson Prentice Hall, 2006, ch. 7-6.
- [2] U. Tietze and C. Schenk, *Halbleiterschaltungstechnik*, 11. Auflage. Springer, 1999, Kap. 6.5.
- [3] C. Thorrold, *Guide for Writing Lab Reports*, Version 1.0.
- [4] Texas Instruments, *NE555 Precision Timers – Datasheet*, <https://www.ti.com/lit/ds/symlink/ne555.pdf>, accessed April 2026.
- [5] Texas Instruments, *CD74C00 Quad 2-Input NAND Gate – Datasheet*, <https://www.ti.com/lit/ds/symlink/cd74hc00.pdf>, accessed April 2026.

A Appendix B – Actual Component Values

Table 6: Component values measured with multimeter before assembly.

Component	Nominal	Measured	Tolerance	Task
R_1 (monostable)	1 M Ω	0.983 M Ω	$\pm 5\%$	3.1
C (monostable)	1 μ F	1.074 μ F	$\pm 20\%$	3.1
R_1 (astable)	10 k Ω	9.87 k Ω	$\pm 5\%$	3.2
R_2 (astable)	10 k Ω	10.12 k Ω	$\pm 5\%$	3.2
C (astable)	100 nF	96.3 nF	$\pm 20\%$	3.2
R_S (inverter)	1 k Ω	0.991 k Ω	$\pm 5\%$	3.3
R (inverter)	10 k Ω	10.08 k Ω	$\pm 5\%$	3.3
C (inverter)	1 nF	0.963 nF	$\pm 20\%$	3.3